Chapter 7. Function Spaces

The space C(X), Arzelà–Ascoli, Weierstrass approximation, sequence spaces, improper integrals, spaces of integrable functions

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7.1 The space $C(X)$

Motivation. Thinking of functions as points of a space with a metric gives a geometric language for approximation and convergence. The space $C(X)$ with the sup norm is complete, and the Stone–Weierstrass theorem says polynomials are dense in it.

A **norm** on a real or complex vector space V is a function $\|\cdot\|$ with $\|v\| \geq 0$ (equality iff $v = 0$), $\|\lambda v\| = |\lambda|\|v\|$, and $\|u + v\| \leq \|u\| + \|v\|$; V is then a **normed space**, with distance $\|u - v\|$, and convergence, Cauchy sequences, open and closed sets are defined as in $\mathbb{R}^n$. A normed space in which every Cauchy sequence converges is **complete** (a **Banach space**). For compact $X \subset \mathbb{R}^n$, $C(X)$ is the space of continuous $f : X \to \mathbb{C}$ with $\|f\|_{\sup} = \max_X |f|$; convergence in this norm is uniform convergence.

Theorem 7.1 (7.1.1)

For compact X, every Cauchy sequence in $C(X)$ converges: $C(X)$ is complete.

Proof. A Cauchy sequence in $C(X)$ is uniformly Cauchy, so converges uniformly (Theorem 6.2) to a function that is continuous (Theorem 6.1). ■

The Heine–Borel theorem fails in $C(X)$: the functions $f_n(x) = x^n$ on $[0, 1]$ form a closed bounded set (no subsequence converges uniformly, since the pointwise limit is discontinuous). The missing ingredient is equicontinuity. A family $\mathcal{F} \subset C(X)$ is **equicontinuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \epsilon$ for all $f \in \mathcal{F}$ and $\|x - y\| < \delta$; it is **bounded** if $\sup_{f\in\mathcal{F}} \|f\|_{\sup} < \infty$.

Proposition 7.2 (7.1.2)

If X is compact and (f_n) converges uniformly in $C(X)$, then $\{f_n\}$ is equicontinuous.

Proof. Given ϵ, choose N with $\|f_n - f_N\|_{\sup} < \epsilon/3$ for $n \geq N$ (Theorem 6.2). The finitely many functions $f_1, \dots, f_N$ are uniformly continuous (Theorem 3.18 of Volume I), so some δ works for all of them with $\epsilon/3$. For $n > N$ and $\|x - y\| < \delta$, $|f_n(x) - f_n(y)| \leq |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| < \epsilon$. ■

Theorem 7.3 (7.1.3, Arzelà–Ascoli)

If X is compact and $\mathcal{F} \subset C(X)$ is bounded and equicontinuous, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.

Proof. Let (f_n) be in $\mathcal{F}$, $|f_n| \leq M$. X has a countable dense subset $E = \{p_1, p_2, \dots\}$ (for each k, finitely many balls of radius $1/k$ cover X; take their centres in X). The bounded sequence $(f_n(p_1))$ has a convergent subsequence $(f_{1,n})$; $(f_{1,n}(p_2))$ has a convergent subsequence $(f_{2,n})$; and so on. The diagonal sequence $g_n = f_{n,n}$ is, from the k-th term on, a subsequence of $(f_{k,n})$, so $(g_n(p_k))$ converges for every k. Given ϵ, take δ from equicontinuity for $\epsilon/3$, and finitely many $p_{k_1}, \dots, p_{k_r}$ with every $x \in X$ within δ of one of them. Choose N with $|g_m(p_{k_j}) - g_n(p_{k_j})| < \epsilon/3$ for $m, n \geq N$ and all j. For any x, pick $p = p_{k_j}$ with $\|x - p\| < \delta$; then $|g_m(x) - g_n(x)| \leq |g_m(x) - g_m(p)| + |g_m(p) - g_n(p)| + |g_n(p) - g_n(x)| < \epsilon$. So (g_n) is uniformly Cauchy and converges in $C(X)$ (Theorem 7.1). ■

Combined with Proposition 7.2: a subset of $C(X)$ has compact closure iff it is bounded and equicontinuous.

Theorem 7.4 (7.1.4, Weierstrass approximation)

For every continuous $f : [a, b] \to \mathbb{R}$ there are polynomials $P_n \to f$ uniformly on $[a, b]$.

Proof. By an affine change of variable take $[a, b] = [0, 1]$, and subtracting the linear function $f(0) + x(f(1) - f(0))$ assume $f(0) = f(1) = 0$. Extend f by 0 outside $[0, 1]$; it is uniformly continuous on $\mathbb{R}$. Let $Q_n(x) = c_n(1 - x^2)^n$ on $[-1, 1]$ with c_n chosen so that $\int_{-1}^{1} Q_n = 1$. Since $(1 - x^2)^n \geq 1 - nx^2$,

$$\int_{-1}^{1} (1 - x^2)^n dx \geq 2 \int_{0}^{1/\sqrt{n}} (1 - nx^2) dx = \frac{4}{3\sqrt{n}} > \frac{1}{\sqrt{n}},$$

so $c_n < \sqrt{n}$, and for $\delta \leq |x| \leq 1$, $Q_n(x) \leq \sqrt{n}(1 - \delta^2)^n \to 0$ uniformly. Let

$$P_n(x) = \int_{-1}^{1} f(x + t) Q_n(t)\, dt = \int_{0}^{1} f(t) Q_n(t - x)\, dt \qquad (0 \leq x \leq 1),$$

a polynomial in x. Given ϵ, choose δ with $|f(y) - f(x)| < \epsilon/2$ for $|y - x| < \delta$, and let $M = \max|f|$. Then

$$|P_n(x) - f(x)| \leq \int_{-1}^{1} |f(x + t) - f(x)| Q_n(t)\, dt \leq 2M \cdot 2\sqrt{n}(1 - \delta^2)^n + \frac{\epsilon}{2} < \epsilon$$

for large n, uniformly in $x \in [0, 1]$. ■

For complex-valued f, approximate real and imaginary parts. Consequently $C[a, b]$ has a countable dense set (polynomials with rational coefficients).

7.2 Sequence spaces

Motivation. The sequence spaces ℓ^p are the simplest infinite-dimensional normed spaces. Hölder's and Minkowski's inequalities make them normed and complete, and they are models for the function spaces of Chapter 10.

$\ell^1 = \ell^1(\mathbb{N})$ is the space of sequences $a = (a(n))$ with $\|a\|_1 = \sum|a(n)| < \infty$, and ℓ^2 those with $\|a\|_2 = (\sum|a(n)|^2)^{1/2} < \infty$. ℓ^2 is a vector space with inner product $\langle a, b\rangle = \sum a(n)\overline{b(n)}$, which converges absolutely since $|a(n)b(n)| \leq \frac{1}{2}(|a(n)|^2 + |b(n)|^2)$; $\|\cdot\|_2$ is the associated norm (Theorem 6.17).

Theorem 7.5 (7.2.1)

ℓ^1 is complete.

Proof. Let (a_m) be Cauchy. Since $|a_m(n) - a_k(n)| \leq \|a_m - a_k\|_1$, each $(a_m(n))_m$ converges to some $a(n)$. Given ϵ, choose N with $\|a_m - a_k\|_1 < \epsilon$ for $m, k \geq N$. For each L, $\sum_{n \leq L}|a_m(n) - a_k(n)| < \epsilon$; let $k \to \infty$ and then $L \to \infty$: $\sum_n|a_m(n) - a(n)| \leq \epsilon$ for $m \geq N$. So $a = a_m - (a_m - a) \in \ell^1$ and $a_m \to a$. ■

Theorem 7.6 (7.2.2)

ℓ^2 is complete.

Proof. As in Theorem 7.5: the coordinates converge, and $\sum_{n \leq L}|a_m(n) - a_k(n)|^2 < \epsilon^2$ passes to the limit in k and then in L. ■

The same proof shows that ℓ^∞ (bounded sequences with the sup norm) is complete. The unit vectors e_n in ℓ^2 form a bounded sequence with $\|e_m - e_n\|_2 = \sqrt{2}$, so without convergent subsequence: closed bounded sets in infinite-dimensional spaces need not be compact.

7.3 Improper integrals

Motivation. Many important integrals, such as $\int_0^\infty e^{-x^2}dx$, run over infinite intervals or involve unbounded integrands. Improper integrals extend Riemann integration to them, with convergence tests parallel to those for series.

Let I be an interval and $f : I \to \mathbb{C}$ Riemann integrable on every compact subinterval. If $I = [a, b)$ ($b \le \infty$), the **improper integral** $\int_I f = \lim_{c\to b^-} \int_a^c f$ when the limit exists; similarly at a left end point, and for open intervals both ends are treated separately. We write $\int_I |f| < \infty$ when this improper integral converges (**absolute convergence**).

Proposition 7.7 (7.3.1)

If $0 \le f \le g$ on I and $\int_I g$ converges, then $\int_I f$ converges.

Proof. $c \mapsto \int_a^c f$ is increasing and bounded by $\int_I g$, so it has a limit as $c \to b^-$ (as for monotone sequences). ∎

Proposition 7.8 (7.3.2, comparison test)

If $|f| \le g$ and $\int_I g$ converges, then $\int_I |f|$ and $\int_I f$ converge and $|\int_I f| \le \int_I |f| \le \int_I g$.

Proof. $\int_I |f|$ converges by Proposition 7.7. Cauchy criterion: for $a < c < c' < b$, $|\int_c^{c'} f| \le \int_c^{c'} |f| \to 0$ as $c, c' \to b$, so $\lim \int_a^c f$ exists (as in Theorem 2.14 of Volume I). The inequality passes to the limit from finite intervals (Theorem 5.5 of Volume I). ∎

Example 7.1

$\int_1^\infty x^{-p}dx < \infty$ iff $p > 1$, and $\int_0^1 x^{-p}dx < \infty$ iff $p < 1$. $\int_0^\infty \frac{\sin x}{x}\,dx$ converges (integrate by parts: $\int_1^c \frac{\sin x}{x}dx = \frac{\cos 1}{1} - \frac{\cos c}{c} - \int_1^c \frac{\cos x}{x^2}dx$) but not absolutely (on $[k\pi, (k+1)\pi]$, $\int |\sin x|/x \ge \frac{2}{(k+1)\pi}$).

Proposition 7.9 (7.3.3, integral test)

If $f : [1, \infty) \to (0, \infty)$ is decreasing, then $\sum f(n)$ and $\int_1^\infty f$ converge or diverge together; indeed $\sum_{n=2}^N f(n) \le \int_1^N f \le \sum_{n=1}^{N-1} f(n)$, and if $f \to 0$, then $\sum_{n\le N} f(n) - \int_1^N f$ converges.

Proof. On $[n, n+1]$, $f(n+1) \le f \le f(n)$; integrate and sum. The differences $d_N = \sum_{n\le N} f(n) - \int_1^N f$ decrease ($d_N - d_{N+1} = \int_N^{N+1} f - f(N+1) \ge 0$) and are $\ge f(N) > 0$, so converge. ∎

For $f(x) = 1/x$ this gives Euler's constant $\gamma = \lim(1 + \frac{1}{2} + \cdots + \frac{1}{N} - \log N)$.

Theorem 7.10 (7.3.4)

If $\int_I |f| < \infty$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $\int_x^y |f| < \epsilon$ whenever $x < y$ in I and $y - x < \delta$.

Proof. Choose a compact $[c, d] \subset I$ with $\int_{I-[c,d]} |f| < \epsilon/2$; f is bounded on $[c, d]$, say by K. With $\delta = \epsilon/(2K + 1)$, $\int_x^y |f| \le \int_{[x,y]\cap[c,d]} |f| + \int_{I-[c,d]} |f| < K\delta + \epsilon/2 < \epsilon$. ∎

7.4 Spaces of integrable functions

Motivation. Just as ℓ^p measures sequences, the integral defines norms on functions. These spaces are not complete with the Riemann integral, a defect that motivates the Lebesgue integral of Chapter 10.

Proposition 7.11 (7.4.1)

For a normed space X the following are equivalent:
(a) every Cauchy sequence in X converges;
(b) whenever $\sum\|x_n\| < \infty$, the series $\sum x_n$ converges in X.

Proof. (a)⇒(b): the partial sums are Cauchy, since $\|\sum_{m<k\le n} x_k\| \le \sum_{m<k\le n}\|x_k\|$.

(b)⇒(a): given a Cauchy (y_n), choose $n_1 < n_2 < \cdots$ with $\|y_m - y_n\| < 2^{-k}$ for $m, n \ge n_k$. With $x_1 = y_{n_1}$, $x_{k+1} = y_{n_{k+1}} - y_{n_k}$, $\sum\|x_k\| < \infty$, so $y_{n_k} = \sum_{j\le k} x_j$ converges; a Cauchy sequence with a convergent subsequence converges (Lemma 2.13 of Volume I, same proof). ∎

Let $\mathcal{R}^1(I)$ and $\mathcal{R}^2(I)$ be the locally Riemann integrable $f : I \to \mathbb{C}$ with $\int_I|f| < \infty$, resp. $\int_I|f|^2 < \infty$, with $\|f\|_1 = \int_I|f|$ and $\|f\|_2 = (\int_I|f|^2)^{1/2}$. These are seminorms ($\|f\| = 0$ does not force $f = 0$), and $\mathcal{R}^2$ carries the inner product $\langle f, g\rangle = \int_I f\bar{g}$. Neither space is complete; this is the reason for introducing the Lebesgue integral (Chapter 10). A **step function** is a finite linear combination of characteristic functions of bounded intervals.

Lemma 7.12 (7.4.2)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a step function s with $\|f - s\|_2 < \epsilon$.

Proof. Treating real and imaginary parts separately, let f be real. Choose a compact $[a, b] \subset I$ with $\int_{I-[a,b]}|f|^2 < \epsilon^2/4$. On $[a, b]$, f is Riemann integrable, so bounded by some K; choose a partition with $U(f, P) - L(f, P) < \epsilon^2/(8K)$ and let $s = m_i$ on the i-th interval (and 0 off $[a, b]$). Then $0 \le f - s \le 2K$ on $[a, b]$, so $\int_a^b|f - s|^2 \le 2K\int_a^b(f - s) \le 2K(U - L) < \epsilon^2/4$. Altogether $\|f - s\|_2^2 < \epsilon^2/2$. ∎

Lemma 7.13 (7.4.3)

For a step function s on I and $\epsilon > 0$ there is a continuous g with $\|s - g\|_2 < \epsilon$ (and $g = 0$ outside a compact subinterval).

Proof. It suffices to treat $s = \chi_{[a,b]}$ (with any type of end points). Let g be 1 on $[a + \eta, b - \eta]$, 0 outside (a, b), and linear in between; then $|s - g| \le 1$ and differs from 0 only on two intervals of length η, so $\|s - g\|_2^2 \le 2\eta$. ∎

Theorem 7.14 (7.4.4)

For $f \in \mathcal{R}^2(I)$ and $\epsilon > 0$ there is a continuous g with $\|f - g\|_2 < \epsilon$.

Proof. Combine Lemmas 7.12 and 7.13 with the triangle inequality for $\|\cdot\|_2$. ∎

Corollary 7.15 (7.4.5)

For $f \in \mathcal{R}^2[a, b]$ there are polynomials P_n with $\|f - P_n\|_2 \to 0$.

Proof. Choose continuous g with $\|f-g\|_2<\epsilon$ and a polynomial P with $\|g-P\|_{\sup}<\epsilon$ (Theorem 7.4); then $\|g-P\|_2\le\epsilon\sqrt{b-a}$. ∎

Theorem 7.16 (7.4.6)

For $f\in\mathcal{R}^1(I)$ and $\epsilon>0$ there is a step function s with $\|f-s\|_1<\epsilon$, and a continuous g vanishing outside a compact subinterval with $\|f-g\|_1<\epsilon$. For $f\in\mathcal{R}^1[a,b]$ there are polynomials P with $\|f-P\|_1<\epsilon$.

Proof. As in Lemma 7.12, with $\int_a^b|f-s|=\int_a^b(f-s)\le U-L$ for the lower step function. For the continuous approximation of $\chi_{[a,b]}$ the error in $\|\cdot\|_1$ is at most η. Polynomials: $\|g-P\|_1\le(b-a)\|g-P\|_{\sup}$. ∎

7.5 Selected exercises

Motivation. These problems from the textbook's exercise set use the Weierstrass theorem and the geometry of function and sequence spaces: moments determine a continuous function, Bernstein polynomials give a constructive proof of the Weierstrass theorem, and compactness in ℓ^2 depends on how fast the coordinates are forced to decay.

Exercise 7.5.1 (Moments determine a function)

Let $f:[0,1]\to\mathbb{R}$ be continuous with $\int_0^1 f(x)x^n\,dx=0$ for $n=0,1,2,\dots$. Show that $f=0$.

Solution.

Step 1. By linearity, $\int_0^1 fP=0$ for every polynomial P.

Step 2. By Theorem 7.4 there are polynomials $P_n\to f$ uniformly on $[0,1]$. Then

$$\int_0^1 f^2=\int_0^1 f(f-P_n)\le\|f\|_{\sup}\|f-P_n\|_{\sup}\to 0,$$

so $\int_0^1 f^2=0$.

Step 3. If $f(x_0)\neq 0$, then by continuity $f^2>\tfrac12 f(x_0)^2$ on some interval $J\subset[0,1]$ of length $\eta>0$, and $\int_0^1 f^2\ge\tfrac12 f(x_0)^2\eta>0$, a contradiction. Hence $f=0$. ∎

Exercise 7.5.5 (Weights concentrating at an end point)

Let $f:[0,1]\to\mathbb{R}$ be continuous. Show that

$$\lim_{n\to\infty}\frac{\int_0^1 x^n f(x)\,dx}{\int_0^1 x^n\,dx}=f(1).$$

Solution.

Step 1. Since $\int_0^1 x^n dx=\frac{1}{n+1}$, the quotient minus $f(1)$ equals $(n+1)\int_0^1 x^n(f(x)-f(1))dx$.

Step 2. Let $M=\max|f|$ and $\epsilon>0$. By continuity at 1 choose $\delta\in(0,1)$ with $|f(x)-f(1)|<\epsilon$ for $1-\delta\le x\le 1$. Then

$$(n+1)\int_0^{1-\delta}x^n|f(x)-f(1)|\,dx\le 2M(1-\delta)^{n+1},$$

$$(n+1)\int_{1-\delta}^{1} x^n |f(x)-f(1)|\,dx \le \epsilon\,(n+1)\int_0^1 x^n dx = \epsilon.$$

Step 3. The first bound tends to 0, so the absolute value in Step 1 is less than 2ϵ for large n. As ϵ is arbitrary, the limit is $f(1)$. ■

Exercise 7.5.7 (Bernstein polynomials)

Let $f:[0,1]\to\mathbb{R}$ be continuous, $r_k(x)=\binom{n}{k}x^k(1-x)^{n-k}$, and $p_n(x)=\sum_{k=0}^n \binom{n}{k} f\left(\frac{k}{n}\right)x^k(1-x)^{n-k}$.
(a) Show that $\sum_{k=0}^n (k-nx)^2 r_k(x) = nx(1-x)$.
(b) Show that $p_n \to f$ uniformly on $[0,1]$.

Solution.

(a)

Step 1. By the binomial theorem, $\sum_k r_k(x) = (x+(1-x))^n = 1$.

Step 2. From $k\binom{n}{k} = n\binom{n-1}{k-1}$ and $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$, the binomial theorem again gives

$$\sum_k k\,r_k(x) = nx, \qquad \sum_k k(k-1)\,r_k(x) = n(n-1)x^2.$$

Step 3. Expanding $(k-nx)^2 = k(k-1)+k-2nxk+n^2x^2$,

$$\sum_k (k-nx)^2 r_k(x) = n(n-1)x^2 + nx - 2n^2x^2 + n^2x^2 = nx(1-x).$$

(b)

Step 1. By (a), Step 1, $f(x)-p_n(x) = \sum_k \left(f(x)-f\left(\frac{k}{n}\right)\right) r_k(x)$. Let $M=\max|f|$ and $\epsilon>0$; by uniform continuity choose $\delta>0$ with $|f(x)-f(y)|<\epsilon$ whenever $|x-y|<\delta$.

Step 2. The indices with $\left|\frac{k}{n}-x\right|<\delta$ contribute at most $\epsilon\sum_k r_k(x) = \epsilon$.

Step 3. For the other indices $(k-nx)^2 \ge n^2\delta^2$, so by (a) their contribution is at most

$$2M\sum_k \frac{(k-nx)^2}{n^2\delta^2} r_k(x) = \frac{2Mx(1-x)}{n\delta^2} \le \frac{M}{2n\delta^2}.$$

Step 4. Hence $\|f-p_n\|_{\sup} \le \epsilon + \frac{M}{2n\delta^2} < 2\epsilon$ for large n. This is a constructive proof of Theorem 7.4. ■

Exercise 7.5.9 (Compact and non-compact boxes in ℓ^2)

Show that $K=\{a\in\ell^2 : |a(n)|\le \frac{1}{n},\ n=1,2,\dots\}$ is compact, i.e. every sequence in K has a subsequence converging in ℓ^2 to a point of K. Is $L=\{a\in\ell^2 : |a(n)|\le \frac{1}{\sqrt{n}},\ n=1,2,\dots\}$ compact?

Solution.

(a) K is compact.

Step 1. Let (a_m) be a sequence in K. Each coordinate sequence $(a_m(n))_m$ is bounded by $\frac{1}{n}$. By the diagonal argument of Theorem 7.3, there is a subsequence (b_j) with $b_j(n)\to a(n)$ for every n.

Step 2. $|a(n)|\le\frac{1}{n}$, so $\sum|a(n)|^2 \le \sum n^{-2} < \infty$ and $a\in K$.

Step 3. Given $\epsilon > 0$ choose N with $\sum_{n>N} \frac{4}{n^2} < \frac{\epsilon^2}{2}$, then j_0 with $\sum_{n\le N}|b_j(n)-a(n)|^2 < \frac{\epsilon^2}{2}$ for $j \ge j_0$. Since $|b_j(n)-a(n)| \le \frac{2}{n}$,

$$\|b_j - a\|_2^2 = \sum_{n\le N} |b_j(n)-a(n)|^2 + \sum_{n>N} |b_j(n)-a(n)|^2 < \epsilon^2 \qquad (j \ge j_0).$$

(b) L is not compact.

Step 1. For $m \ge 1$ let $c_m(n) = \frac{1}{\sqrt{n}}$ for $2^m < n \le 2^{m+1}$ and $c_m(n) = 0$ otherwise. Then $c_m \in L$ and

$$\|c_m\|_2^2 = \sum_{2^m < n \le 2^{m+1}} \frac{1}{n} \ge 2^m \cdot \frac{1}{2^{m+1}} = \frac{1}{2}.$$

Step 2. The c_m have disjoint supports, so $\|c_m - c_{m'}\|_2^2 = \|c_m\|_2^2 + \|c_{m'}\|_2^2 \ge 1$ for $m \ne m'$. No subsequence is Cauchy, so none converges. ∎